Library Management System

Stock, issue and return tracking, automated

A low-code application built on Zoho Creator

Thirukumaran T

Register No. 240701568

CB23F35 - Low-Code Application Development using Zoho Creator

Rajalakshmi Engineering College | November 2026

Problem: manual registers drift from reality

1

Stale stock counts

Available copies are updated only when someone remembers.

2

Late overdue detection

Due dates sit on a slip; fines are worked out by hand.

3

No totals or search

Records are scattered, with no instant reports.

4

No safeguards

A book can be issued at zero stock, or a return entered twice.

Solution: one app to record, control and monitor

1 Record

Four linked forms: Book Details, Student Details, Book Issue and Book Return, joined by lookups.

2 Control

Deluge scripts check stock, validate dates and quantities, block duplicate returns and calculate fines.

3 Monitor

Five table reports with search and filters, an Overdue Books report and a plain Admin Dashboard.

deployed

Why low-code on Zoho Creator

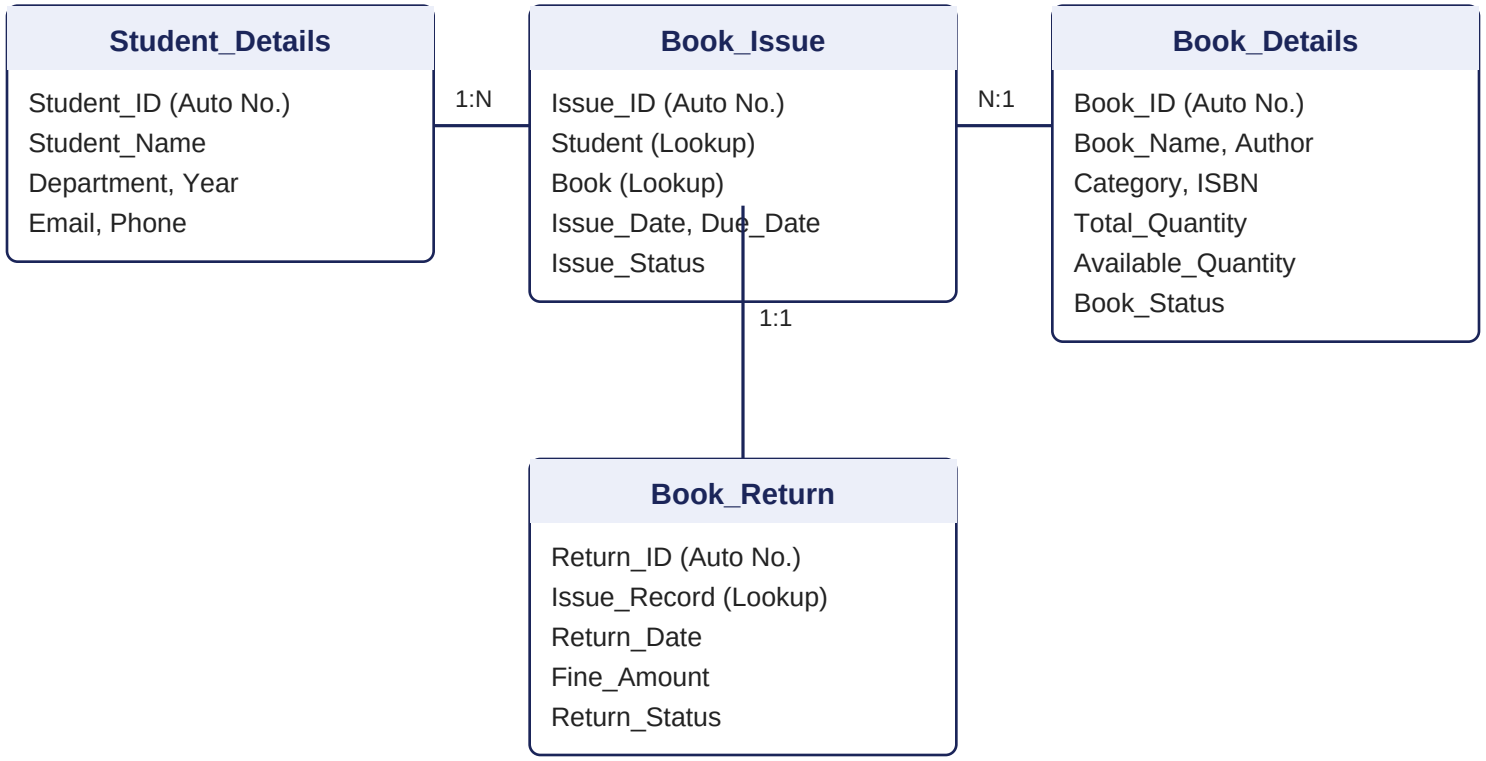
Criterion	Traditional (pro-code)	Zoho Creator
Development time	UI, database and server code written and integrated	Forms and reports built visually; Zia generates a start
Hosting	Own or rented server, backups	Built-in Zoho cloud hosting and backups
Mobile app	Separate app must be developed	Zoho Creator mobile app by default
Security	Login and roles coded by hand	Built-in login, roles and permissions
Change	Code, test, redeploy	Edit a form or script and publish

Trade-off: the interface follows the default look and complex logic must fit Deluge, which suits a records-driven library.

System architecture

Presentation	Web browser (Chrome / Edge / Firefox)	Zoho Creator mobile app	Default menu navigation
Application	4 forms (data capture)	5 list reports	Admin Dashboard page
Logic and automation	5 form workflows	5 Deluge scripts (DS-01 to DS-05)	Validate on submit, update on success
Data and services	Creator cloud database	Zoho Mail sendmail (planned)	Zia AI builder (build time)

Data model: Book Issue links students and books



Four forms, no subforms

- Student and Book are chosen by lookup, never typed.
- Book Return links to exactly one issue record.
- Quantities live on the book record, so reports read one value.
- Issue Status makes the duplicate-return check a single comparison.

The application at a glance

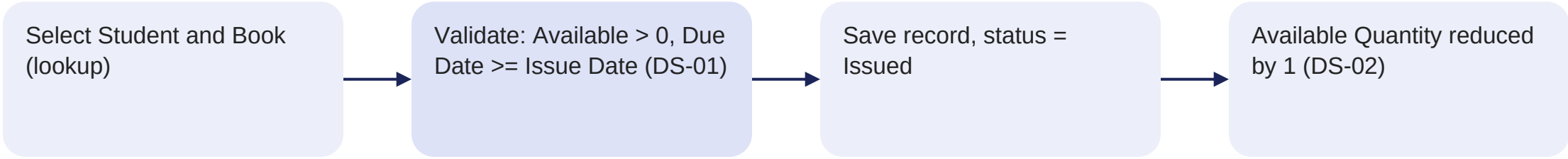
4	5	5	5
Forms	List reports	Form workflows	Deluge scripts

Also included

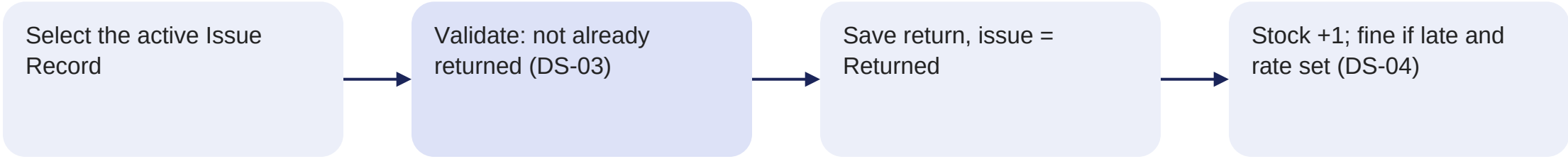
- Admin Dashboard page with four totals and a Recent Transactions table
- Zoho Creator mobile app access without extra development
- Administrator profile with full access; audit trail of every record change

Automation: issue and return workflows

Issue a book



Return a book



Validation runs On Validate and can cancel the submit; stock updates run On Success, after the record is saved.

A little code where it matters

ID	Trigger	Purpose	Concepts
DS-01	Book_Issue, On Validate	Block issue if no copies or bad dates	fetch, if, cancel submit
DS-02	Book_Issue, On Success	Reduce quantity; set Out of Stock	record update
DS-03	Book_Return, On Validate	Block repeat return of same issue	fetch, if, cancel submit
DS-04	Book_Return, On Success	Restore stock, mark Returned, fine	date functions
DS-05	Book_Details, On Validate	Keep quantities valid	if, cancel submit

```
bk = Book_Details[ID == input.Book];
if(bk.Available_Quantity <= 0)
{
    alert "No copies available for " + bk.Book_Name;
    cancel submit;
}
```

DS-01: issue validation

- Each script fetches the related record by ID
- Fines apply only once a per-day rate is approved
- No fetch statements inside loops

Dashboard and reports

Rs. 17.50	5	6	2
Total fine	Total transactions	Total students	Overdue

0	2	1	2
Lost	Returned	Issued	Overdue

Five list reports

- Book Details and Student Details, searchable by name or category
- Book Issue (filter by status) and Book Return with fines

Overdue Books

- Criteria: Issue_Status != Returned and Due_Date < today
- Only active past-due issues appear

Figures from the running application. Kanban views group students by status and books by category.

Testing and results

15

test cases

- Cover every functional requirement
- Negative tests: zero stock, duplicate return, bad dates, bad quantities

Step	Action	Result	Java qty
1	Open dashboard	Totals shown	10
2	Issue Java to Arun Kumar	Issue saved	9
3	Return the book	Returned, stock +1	10
4	Return it again	Blocked: already returned	10
5	Issue at zero stock	Blocked: no copies	0

Sample scenario, checked on the live dashboard.

Limitations, next steps and thanks

Today

No email or SMS reminders. No student portal. Fine rate must be approved before fines are enabled. Plain default interface; trial edition.

Next

Scheduled workflow for overdue reminders, student login portal, barcode scanning, online fine payment and Zoho Analytics.

Thank you

Questions and live demo